Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2022 – Paper 2 – Solutions**

June 2022 – Question 1 – Solution

(a) (i) Let p and q be any two propositions. Complete the truth table below. [3]

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim\boldsymbol{q}$	$(\sim\boldsymbol{q} \wedge \boldsymbol{p})$	$\boldsymbol{p} \vee (\sim\boldsymbol{q} \wedge \boldsymbol{p})$
T	T	F	F	T
T	F	T	T	T
F	T	F	F	F
F	F	T	F	F

(ii) Hence, state whether the statements $(\sim\boldsymbol{q} \wedge \boldsymbol{p})$ and $\boldsymbol{p} \vee (\sim\boldsymbol{q} \wedge \boldsymbol{p})$ are logically equivalent. Justify your response. [2]

$(\sim\boldsymbol{q} \wedge \boldsymbol{p})$ and $\boldsymbol{p} \vee (\sim\boldsymbol{q} \wedge \boldsymbol{p})$ are not logically equivalent because they do not have the same truth values according to the last two columns in the truth table above.

(b) Determine the values of a, b, and c, such that

$$2x^2 - 7x + 12 = a(x-2)(x-1) - b(x-3) + c.$$ [9]

$$2x^2 - 7x + 12 = a(x-2)(x-1) - b(x-3) + c$$

$$2x^2 - 7x + 12 = a(x^2 - 3x + 2) - bx + 3b + c$$

$$2x^2 - 7x + 12 = ax^2 - 3ax + 2a - bx + 3b + c$$

$$2x^2 - 7x + 12 = ax^2 - (3a + b)x + (2a + 3b + c)$$

Comparing and equating corresponding coefficients gives:

$a = 2$ → Equation 1

$3a + b = 7$ → Equation 2

$2a + 3b + c = 12$ → Equation 3

Substituting Equation 1 into Equation 2 gives:

$$3(2) + b = 7$$
$$6 + b = 7$$
$$b = 7 - 6$$
$$b = 1$$

Substituting $a = 2$ and $b = 1$ into Equation 3 gives:

$$2(2) + 3(1) + c = 12$$
$$4 + 3 + c = 12$$
$$7 + c = 12$$
$$c = 12 - 7$$
$$c = 5$$

$\therefore a = 2, b = 1, c = 5$

(c) Solve the inequality $|3x + 2| \geq 4$. [5]

<u>Case 1:</u>

$3x + 2 \geq 4$

$3x \geq 4 - 2$

$3x \geq 2$

$x \geq \frac{2}{3}$

<u>Case 2:</u>

$-(3x + 2) \geq 4$

$3x + 2 \leq -4$

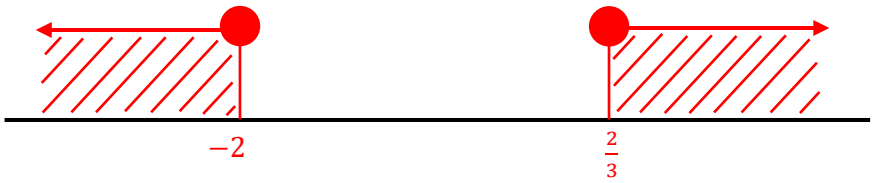
$3x \leq -4 - 2$

$3x \leq -6$

$x \leq \frac{-6}{3}$

$x \leq -2$

So, we have $x \geq \frac{2}{3}$ and $x \leq -2$.



$\therefore$ The solution is $\left\{x: x \leq -2 \cup x \geq \frac{2}{3}\right\}$.

(d) Solve the logarithmic equation $\log_5(x+2) + \log_5(x+6) = 1$. [6]

$\log_5(x+2) + \log_5(x+6) = 1$

$\log_5(x+2)(x+6) = 1$

Converting to exponential form gives:

$(x+2)(x+6) = 5^1$

$x^2 + 2x + 6x + 12 = 5$

$x^2 + 8x + 7 = 0$

$(x+7)(x+1) = 0$

Either $x + 7 = 0$ or $x + 1 = 0$

$x = -7$ $x = -1$

(invalid)

$\therefore x = -1$

Total: 25 marks

June 2022 – Question 2 – Solution

(a) Let $f(x) = \frac{3x+1}{x}$ and $g(x) = e^{-2x} + 1$.

(i) Show that $f^{-1}(x) = \frac{1}{x-3}$. [3]

$f(x) = \frac{3x+1}{x}$

Let $y = f(x)$.

$y = \frac{3x+1}{x}$

Interchanging variables x and y:

$x = \frac{3y+1}{y}$

Making y the subject of the formula:

$$xy = 3y + 1$$

$$xy - 3y = 1$$

$$y(x-3) = 1$$

$$y = \frac{1}{x-3}$$

$\therefore f^{-1}(x) = \frac{1}{x-3}$

Q.E.D.

(ii) Hence, or otherwise, write an expression for $(f^{-1} \circ g)(x)$. [2]

$f^{-1}(x) = \frac{1}{x-3}$ and $g(x) = e^{-2x} + 1$

$$(f^{-1} \circ g)(x) = f^{-1}[g(x)]$$
$$= f^{-1}(e^{-2x} + 1)$$
$$= \frac{1}{(e^{-2x}+1)-3}$$
$$= \frac{1}{e^{-2x}-2}$$

$$\therefore (f^{-1} \circ g)(x) = \frac{1}{e^{-2x}-2}$$

(b) Solve the equation $6 - \frac{7}{2^{2x}} - \frac{3}{4^{2x}} = 0$. [8]

$$6 - \frac{7}{2^{2x}} - \frac{3}{4^{2x}} = 0$$

$$6 - \frac{7}{(2^{2x})} - \frac{3}{(2^{2x})^2} = 0$$

Let $u = 2^{2x}$.

Then, we have,

$$6 - \frac{7}{u} - \frac{3}{u^2} = 0$$

$(\times u^2)$

$$6u^2 - 7u - 3 = 0$$

$$6u^2 - 9u + 2u - 3 = 0$$

$$3u(2u - 3) + 1(2u - 3) = 0$$

$$(3u + 1)(2u - 3) = 0$$

Either $3u + 1 = 0$ or $2u - 3 = 0$

$3u = -1$ $\quad$ $2u = 3$

$u = -\frac{1}{3}$ $\quad$ $u = \frac{3}{2}$

When $u = -\frac{1}{3}$,

$$2^{2x} = -\frac{1}{3}$$

Taking logs on both sides:

$$\log 2^{2x} = \log\left(-\frac{1}{3}\right)$$

$\Rightarrow$ No real solutions exist for x

When $u = \frac{3}{2}$,

$$2^{2x} = \frac{3}{2}$$

Taking logs on both sides:

$\log 2^{2x} = \log\left(\frac{3}{2}\right)$

$2x \log 2 = \log\left(\frac{3}{2}\right)$

$x = \frac{\log\left(\frac{3}{2}\right)}{2\log 2}$

$x = \frac{\log\left(\frac{3}{2}\right)}{\log 2^2}$

$x = \frac{\log\left(\frac{3}{2}\right)}{\log 4}$

$\therefore x = \frac{\log\left(\frac{3}{2}\right)}{\log 4}$

(c) The function $f(x) = 2x^3 - px^2 + qx - 10$ is divisible by $x - 1$ and has a remainder of -6 when divided by $x + 1$.

Calculate the values of p and q. [6]

$f(x) = 2x^3 - px^2 + qx - 10$

Since $f(x)$ is divisible by $(x - 1)$, then by the Factor Theorem, $f(1) = 0$.

$f(1) = 2(1)^3 - p(1)^2 + q(1) - 10$

$0 = 2 - p + q - 10$

$0 = -8 - p + q$

$q = p + 8$ $\rightarrow$ Equation 1

Since $f(x)$ has a remainder of -6 when divided by $(x+1)$, then by the Remainder Theorem, $f(-1) = -6$.

$f(-1) = 2(-1)^3 - p(-1)^2 + q(-1) - 10$

$-6 = -2 - p - q - 10$

$-6 = -12 - p - q$

$q = -12 + 6 - p$

$q = -6 - p$ $\rightarrow$ Equation 2

Equating Equation 1 and Equation 2 gives:

$p + 8 = -6 - p$

$p + p = -6 - 8$

$2p = -14$

$p = \frac{-14}{2}$

$p = -7$

Substituting $p = -7$ into Equation 1 gives:

$q = -7 + 8$

$= 1$

$\therefore p = -7$ and $q = 1$

(d) The roots of the equation $2x^3 - x^2 + 3x - 1 = 0$ are α, β, and γ.

Given that $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{3}{2}$ and $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\alpha+\beta+\gamma}{\alpha\beta\gamma}$, determine the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$. [6]

$2x^3 - x^2 + 3x - 1 = 0$

which is in the form $ax^3 + bx^2 + cx + d = 0$,

where $a = 2, b = -1, c = 3$ and $d = -1$.

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= \frac{-(-1)}{2}$$

$$= \frac{1}{2}$$

We are given that $\alpha\beta + \beta\gamma + \alpha\gamma = \frac{3}{2}$.

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= \frac{-(-1)}{2}$$

$$= \frac{1}{2}$$

For the new roots:

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma+\alpha\gamma+\alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{3}{2} \div \frac{1}{2}$$

$$= \frac{3}{2} \times \frac{2}{1}$$

$$= 3$$

$$\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right)+\left(\frac{1}{\beta}\right)\left(\frac{1}{\gamma}\right)+\left(\frac{1}{\alpha}\right)\left(\frac{1}{\gamma}\right)=\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\alpha\gamma}$$

$$=\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma}$$

$$=\frac{1}{2}\div\frac{1}{2}$$

$$=1$$

$$\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right)\left(\frac{1}{\gamma}\right)=\frac{1}{\alpha\beta\gamma}$$

$$=1\div\frac{1}{2}$$

$$=1\times 2$$

$$=2$$

$\therefore$ The new equation is: $x^3-3x^2+x-2=0$

Total: 25 marks

June 2022 – Question 3 – Solution

(a) Prove that

$$\frac{\tan\theta\sin\theta}{1-\cos\theta} = 1 + \frac{1}{\cos\theta}.$$ [6]

Taking L.H.S:

$$\frac{\tan\theta\sin\theta}{1-\cos\theta} = \frac{\left(\frac{\sin\theta}{\cos\theta}\right)\sin\theta}{1-\cos\theta}$$

$$= \frac{\left(\frac{\sin^2\theta}{\cos\theta}\right)}{1-\cos\theta}$$

$$= \frac{\sin^2\theta}{\cos\theta} \div (1-\cos\theta)$$

$$= \frac{\sin^2\theta}{\cos\theta} \times \frac{1}{1-\cos\theta}$$

$$= \frac{1-\cos^2\theta}{\cos\theta} \times \frac{1}{1-\cos\theta}$$

$$= \frac{(1-\cos\theta)(1+\cos\theta)}{\cos\theta} \times \frac{1}{1-\cos\theta}$$

$$= \frac{1+\cos\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta} + \frac{\cos\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta} + 1$$

$$= 1 + \frac{1}{\cos\theta}$$

$$= \text{R.H.S.}$$

$$\therefore \frac{\tan\theta\sin\theta}{1-\cos\theta} = 1 + \frac{1}{\cos\theta}$$

Q.E.D.

(b) Solve the equation $\tan^2\theta - 2\tan\theta = 3$ for $0 \le \theta \le 2\pi$. [6]

$$\tan^2\theta - 2\tan\theta = 3$$

$$\tan^2\theta - 2\tan\theta - 3 = 0$$

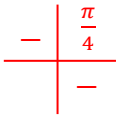
$$(\tan\theta + 1)(\tan\theta - 3) = 0$$

Either $\tan\theta + 1 = 0$ or $\tan\theta - 3 = 0$.

Consider: $\tan\theta + 1 = 0$

$$\tan\theta = -1$$

$$\theta = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

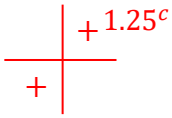


$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Consider: $\tan\theta - 3 = 0$

$$\tan\theta = 3$$

$$\theta = 1.25^c, \pi + 1.25^c$$



$$\theta = 1.5^c, 4.39^c$$

$$\therefore \theta = 1.25^c, \frac{3\pi}{4}, 4.39^c, \frac{7\pi}{4}$$

(c) (i) Show that $4\cos\theta + 3\sin\theta = 5\sin(\theta + 0.927^c)$. [5]

Let $4\cos\theta + 3\sin\theta = r\sin(\theta + \alpha)$

$4\cos\theta + 3\sin\theta = r\cos\theta\sin\alpha + r\sin\theta\cos\alpha$

Equating coefficients gives:

$r\sin\alpha = 4$ $\rightarrow$ Eq. 1

$r\cos\alpha = 3$ $\rightarrow$ Eq. 2

Eq. 1 $\div$ Eq. 2 gives,

$\frac{r\sin\alpha}{r\cos\alpha} = \frac{4}{3}$

$\tan\alpha = \frac{4}{3}$

$\alpha = 0.927^c$

$r = \sqrt{4^2 + 3^2}$

$= \sqrt{25}$

$= 5$

$\therefore 4\cos\theta + 3\sin\theta = 5\sin(\theta + 0.927^c)$

Q.E.D.

(ii) Hence, or otherwise, solve the equation $4\cos\theta + 3\sin\theta = 0$. [4]

$$4\cos\theta + 3\sin\theta = 0$$

$$5\sin(\theta + 0.927^c) = 0$$

$$\sin(\theta + 0.927^c) = 0$$

$$\theta + 0.927^c = 0, \pi, 2\pi$$

$$\theta = 0 - 0.927^c, \pi - 0.927^c, 2\pi - 0.927^c$$

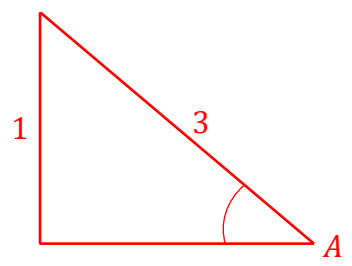
$$\theta = -0.927^c, 2.21^c, 5.36^c$$

$$\therefore \theta = 2.21^c, 5.36^c \quad \text{for } 0 \le \theta \le 2\pi$$

(d) Given that $\sin A = \frac{1}{3}$ and A is obtuse, calculate the value of $\cos A$ without using a calculator. [4]

$$\sin A = \frac{1}{3}$$

$$\sin A = \frac{opp}{hyp}$$



By Pythagoras' Theorem,

$$a^2 + b^2 = c^2$$

$$(1)^2 + b^2 = (3)^2$$

$$1 + b^2 = 9$$

$$b^2 = 9 - 1$$

$$b^2 = 8$$

$$b = \sqrt{8}$$

Since A is obtuse, it lies in the second quadrant.

In the second quadrant, $\cos A$ is negative.

Now,

$$\cos A = \frac{adj}{hyp}$$

$$= -\frac{\sqrt{8}}{3}$$

$$= -\frac{2\sqrt{2}}{3}$$

$$\therefore \cos A = -\frac{2\sqrt{2}}{3}$$

Total: 25 marks

June 2022 – Question 4 – Solution

(a) A circle has equation $x^2 + y^2 - 10x + 4y - 5 = 0$.

(i) Determine the centre and radius of the circle. [4]

$$x^2 + y^2 - 10x + 4y - 5 = 0$$

$$x^2 - 10x + y^2 + 4y = 5$$

$$x^2 - 10x + 25 + y^2 + 4y + 4 = 5 + 25 + 4$$

$$(x-5)^2 + (y+2)^2 = 34$$

which is in the form $(x-a)^2 + (y-b)^2 = r^2$,

where the centre is $(5, -2)$ and the radius, $r = \sqrt{34}$.

(ii) Determine the equation of the tangent to the circle at the point $(2, 3)$. [4]

Consider the points $(5, -2)$ and $(2, 3)$.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3-(-2)}{2-5}$$

$$= -\frac{5}{3}$$

Gradient of normal $= -\frac{5}{3}$

Gradient of tangent $= \frac{3}{5}$

Substituting $m = \frac{3}{5}$ and point $(2, 3)$ into $y - y_1 = m(x - x_1)$ gives:

$$y - 3 = \frac{3}{5}(x - 2)$$

$$y - 3 = \frac{3}{5}x - \frac{6}{5}$$

$$y = \frac{3}{5}x - \frac{6}{5} + 3$$

$$y = \frac{3}{5}x + \frac{9}{5}$$

$$5y = 3x + 9$$

$\therefore$ The equation of the tangent is: $5y = 3x + 9$

(b) (i) Determine the equation of the plane that passes through the point $(0, 2, -2)$ and which is perpendicular to the vector $\boldsymbol{v} = 3\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} + 4\hat{\boldsymbol{k}}$. [3]

Let $\boldsymbol{a} = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$ and $\boldsymbol{n} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$.

The equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} = (0)(3) + (2)(2) + (-2)(4)$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} = 0 + 4 - 8$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} = -4$$

(ii) Determine the angle between the vector $3\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}}$ and the x-axis. [3]

We have the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. The unit vector on the x-axis is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

$$\boldsymbol{a} \cdot \boldsymbol{b} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= (3)(1) + (-2)(0)$$

$$= 3 + 0$$

$$= 3$$

$$|\boldsymbol{a}| = \sqrt{(3)^2 + (-2)^2}$$

$$= \sqrt{19 + 4}$$

$$= \sqrt{13}$$

$$|\boldsymbol{b}| = \sqrt{(1)^2 + (0)^2}$$

$$= \sqrt{1}$$

$$= 1$$

Now,

$$\cos\theta = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|}$$

$$\cos\theta = \frac{3}{(\sqrt{13})(1)}$$

$$\cos\theta = \frac{3}{\sqrt{13}}$$

$$\theta = \cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$$

$$\theta = 33.7° \qquad \text{(to 1 decimal place)}$$

(c) The parametric equations of a line are given as

$$x = 2 + 5\lambda, \quad y = -3 + 4\lambda, \quad z = 4 - 2\lambda.$$

Determine the coordinates of the point where the line crosses the xy plane. [5]

On the xy-plane, $z = 0$.

When $z = 0$,

$$0 = 4 - 2\lambda$$

$$2\lambda = 4$$

$$\lambda = \frac{4}{2}$$

$$\lambda = 2$$

The values of x and y are found as follows:

$x = 2 + 5(2)$

$= 2 + 10$

$= 12$

$y = -3 + 4(2)$

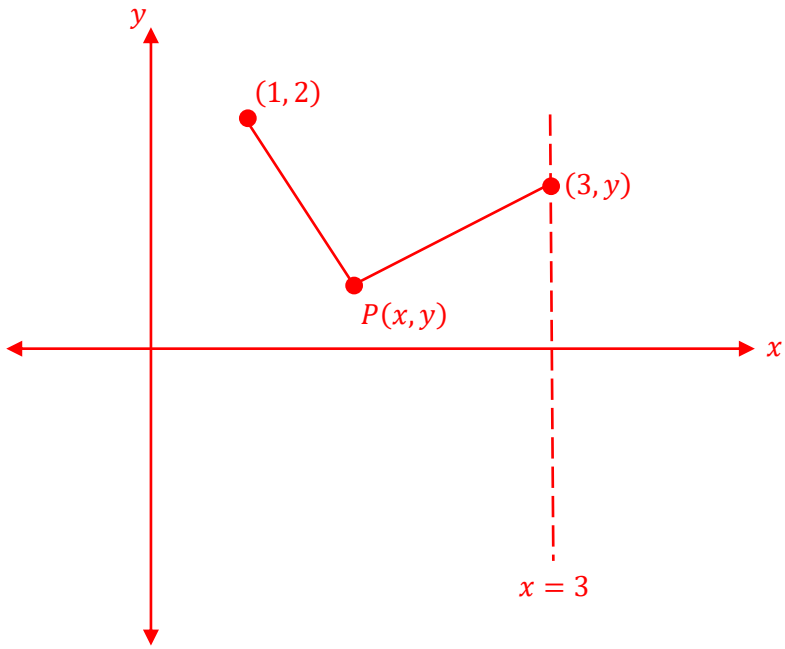
$= -3 + 8$

$= 5$

$\therefore$ The point where the line crosses the xy plane is $(12, 5, 0)$.

(d) A point $P(x, y)$ moves in the xy plane such that it is the same distance from the point $A(1, 2)$ as it is from the line $x = 3$. Determine the equation of the locus of P. [6]

Consider the diagram below:



Consider points $(1, 2)$ and $P(x, y)$.

Distance $= \sqrt{(x-1)^2 + (y-2)^2}$

Consider points $(3, y)$ and $P(x, y)$.

Distance $= \sqrt{(x-3)^2 + (y-y)^2}$

$= \sqrt{(x-3)^2}$

Now,

$$\sqrt{(x-1)^2+(y-2)^2}=\sqrt{(x-3)^2}$$

Squaring both sides gives,

$$(x-1)^2+(y-2)^2=(x-3)^2$$

$$x^2-2x+1+y^2-4y+4=x^2-6x+9$$

$$y^2-2x+6x-4y+1+4-9=0$$

$$y^2+4x-4y-4=0$$

$\therefore$ The equation of the locus of P is: $y^2+4x-4y-4=0$

Total: 25 marks

June 2022 – Question 5 – Solution

(a) Determine $\lim_{x\to 2} \frac{x^2+3x-10}{x^2+x-6}$. [5]

$$\lim_{x\to 2} \frac{x^2+3x-10}{x^2+x-6} = \lim_{x\to 2} \frac{(x-2)(x+5)}{(x-2)(x+3)}$$

$$= \lim_{x\to 2} \frac{x+5}{x+3}$$

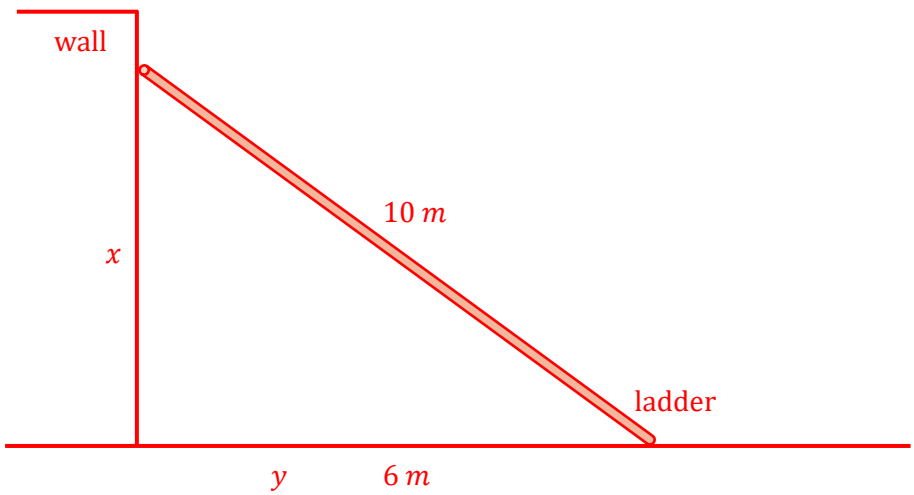
$$= \frac{(2)+5}{(2)+3}$$

$$= \frac{7}{5}$$

$$\therefore \lim_{x\to 2} \frac{x^2+3x-10}{x^2+x-6} = \frac{7}{5}$$

(b) A ladder which is 10 metres long is leaning against a wall. The bottom of the ladder is sliding away from the base of the wall at a rate of 4 m/s. Determine the rate at which the top of the ladder is moving down the wall when the bottom of the ladder is 6 metres from the base of the wall. [7]

Consider the diagram below:



From the question, $\frac{dy}{dt} = 4\ ms^{-1}$.

Using Pythagoras' Theorem,

$x^2 + y^2 = 10^2$

$$x^2 = 10^2 - y^2$$

$$x = \sqrt{100 - y^2}$$

$$x = (100 - y^2)^{\frac{1}{2}}$$

$$\frac{dx}{dy} = \frac{1}{2}(-2y)(100 - y^2)^{\frac{1}{2} - 1}$$

$$\frac{dx}{dy} = -y(100 - y^2)^{-\frac{1}{2}}$$

When $y = 6$,

$\frac{dx}{dy} = -(6)(100 - (6)^2)^{-\frac{1}{2}}$

$$= -(6)(100 - 36)^{-\frac{1}{2}}$$

$$= -(6)(64)^{-\frac{1}{2}}$$

$$= -(6)\left(\frac{1}{8}\right)$$

$$= -\frac{3}{4}$$

Now,

$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt}$

$$= -\frac{3}{4} \times 4$$

$$= -3$$

$\therefore$ The rate at which the top of the ladder is moving down the wall is $3\ ms^{-1}$.

(c) A function f is given as $f(x) = 4x^3 - 3x^2 + 1$, for $-1 \leq x \leq 1$.

(i) Determine the coordinates of the stationary points of the function f. [6]

$f(x) = 4x^3 - 3x^2 + 1$

$f'(x) = 12x^2 - 6x$

At stationary points, $f'(x) = 0$.

When $f'(x) = 0$,

$12x^2 - 6x = 0$

$6x(2x - 1) = 0$

Either $6x = 0$ or $2x - 1 = 0$

$x = 0$ $x = \frac{1}{2}$

When $x = 0$,

$y = 4(0)^3 - 3(0)^2 + 1$

$= 1$

When $x = \frac{1}{2}$,

$y = 4\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 + 1$

$= \frac{3}{4}$

$\therefore$ The stationary points are $(0, 1)$ and $\left(\frac{1}{2}, \frac{3}{4}\right)$.

(ii) Determine the nature of these stationary points. [4]

$f'(x) = 12x^2 - 6x$

$f''(x) = 24x - 6$

When $x = 0$,

$f''(0) = 24(0) - 6$

$= 0 - 6$

$= -6 \quad (< 0)$

So, there is a maximum point when $x = 0$.

When $x = \frac{1}{2}$,

$f''\left(\frac{1}{2}\right) = 24\left(\frac{1}{2}\right) - 6$

$= 12 - 6$

$= 6 \quad (> 0)$

So, there is a minimum point when $x = \frac{1}{2}$.

$\therefore (0, 1)$ is a maximum point and $\left(\frac{1}{2}, \frac{3}{4}\right)$ is a minimum point.

(iii) Determine the **absolute** maximum and minimum values of the function f. [3]

$f(x) = 4x^3 - 3x^2 + 1$, for $-1 \leq x \leq 1$.

The minimum value of the function occurs at $x = -1$.

When $x = -1$,

$f(-1) = 4(-1)^3 - 3(-1)^2 + 1$

$= 4(-1) - 3(1) + 1$

$= -4 - 3 + 1$

$= -6$

Now,

$|f(-1)| = |-6|$

$= 6$

The maximum value of the function occurs at $x = 1$.

When $x = 1$,

$f(1) = 4(1)^3 - 3(1)^2 + 1$

$= 4(1) - 3(1) + 1$

$= 4 - 3 + 1$

$= 2$

Now,

$|f(1)| = |2|$

$= 2$

$\therefore$ The absolute minimum and maximum values of the function f are 6 and 2 respectively.

Total: 25 marks

June 2022 – Question 6 – Solution

(a) Using the substitution $u = x^2 + 1$, determine $\int 4x(x^2+1)^5\,dx$. [5]

$$u = x^2 + 1$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

Now,

$$\int 4x(x^2+1)^5\,dx = 2\int 2x(u)^5\,\frac{du}{2x}$$

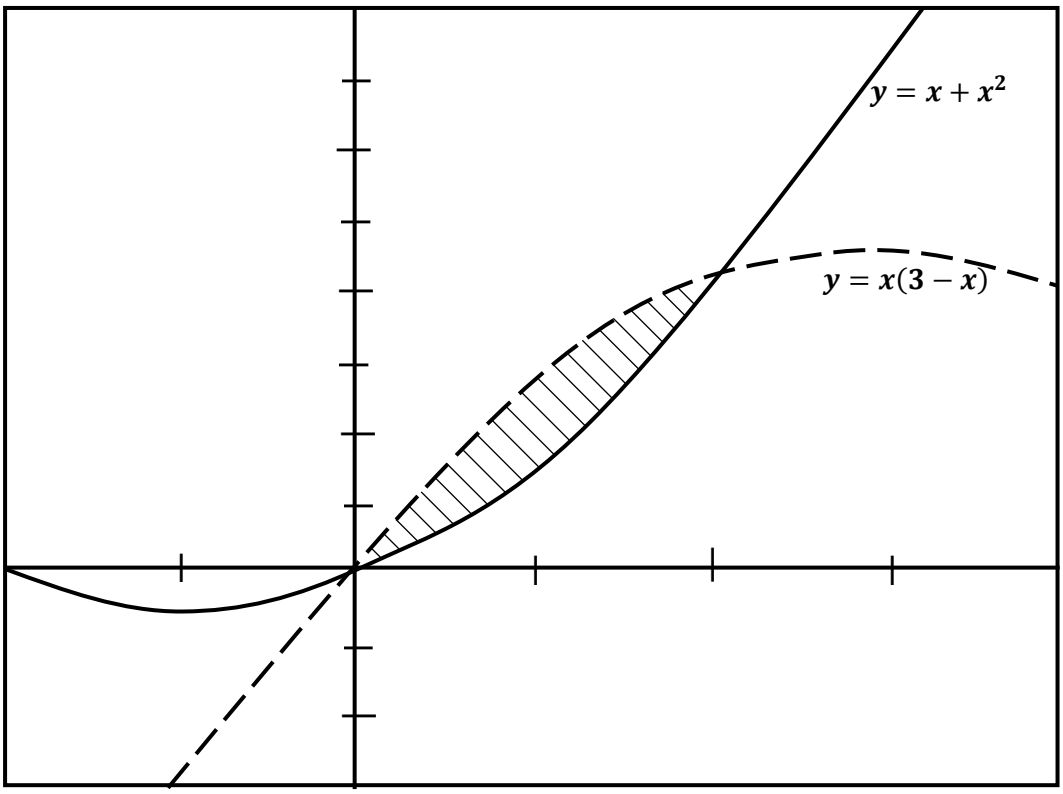
$$= 2\int u^5\,du$$

$$= \frac{2u^6}{6} + c$$

$$= \frac{(x^2+1)^6}{3} + c$$

$$\therefore \int 4x(x^2+1)^5\,dx = \frac{(x^2+1)^6}{3} + c$$

(b) The diagram below shows two curves, $y = x + x^2$ and $y = x(3 - x)$. Calculate the area of the shaded region. [6]



$y = x + x^2$ → Equation 1

$y = x(3 - x)$ → Equation 2

Equating both equations gives:

$$x + x^2 = x(3 - x)$$

$$x + x^2 = 3x - x^2$$

$$x^2 + x^2 + x - 3x = 0$$

$$2x^2 - 2x = 0$$

$$2x(x - 1) = 0$$

Either $2x = 0$ or $x - 1 = 0$

$x = 0$ $x = 1$

The boundaries are $x = 0$ and $x = 1$.

Now,

Area under maximum curve $= \int_a^b y\,dx$

$$= \int_0^1 x(3 - x)\,dx$$

$$= \int_0^1 (3x - x^2)\,dx$$

$$= \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^1$$

$$= \left[\frac{3(1)^2}{2} - \frac{(1)^3}{3}\right] - \left[\frac{7(0)^2}{2} - \frac{(0)^3}{3}\right]$$

$$= \left(\frac{3}{2} - \frac{1}{3}\right) - 0$$

$$= \frac{7}{6}\ units^2$$

And,

Area under minimum curve $= \int_a^b y\,dx$

$$= \int_0^1 (x + x^2)\,dx$$

$$= \left[\frac{x^2}{2} + \frac{x^3}{3}\right]_0^1$$

$$= \left[\frac{(1)^2}{2} + \frac{(1)^3}{3}\right] - \left[\frac{(0)^2}{2} + \frac{(0)^3}{3}\right]$$

$$= \left(\frac{1}{2} + \frac{1}{3}\right) - 0$$

$$= \frac{5}{6}\ units^2$$

Hence,

Area of shaded region = Area under maximum curve − Area under minimum curve

$$= \frac{7}{6} - \frac{5}{6}$$

$$= \frac{1}{3}\ units^2$$

(c) Calculate the volume of the solid generated by revolving the region bounded by the line $y = 6x$ and the parabola $y = 6x^2$ about the x-axis. [6]

Consider,

$y = 6x$ → Equation 1

$y = 6x^2$ → Equation 2

Equating both equations gives:

$$6x = 6x^2$$

$$6x^2 - 6x = 0$$

$$6x(x - 1) = 0$$

Either $6x = 0$ or $x - 1 = 0$

$x = 0$ $x = 1$

Now,

$$y = 6x$$

$$y^2 = (6x)^2$$

$$y^2 = 36x^2$$

So, we have,

$$V_1 = \pi \int_a^b y^2\, dx$$

$$= \pi \int_0^1 (36x^2)\, dx$$

$$= 36\pi \left[\frac{x^3}{3}\right]_0^1$$

$$= 36\pi \left[\frac{(1)^3}{3} - \frac{(0)^3}{3}\right]$$

$$= 12\pi\ units^3$$

Now,

$$y = x^2$$

$$y^2 = (6x^2)^2$$

$$y^2 = 36x^4$$

So, we have,

$$V_2 = \pi \int_a^b y^2\, dx$$

$$= \pi \int_0^1 (36x^4)\, dx$$

$$= 36\pi \left[\frac{x^5}{5}\right]_0^1$$

$$= 36\pi \left[\frac{(1)^5}{5} - \frac{(0)^5}{5}\right]$$

$$= \frac{36\pi}{5}\ units^3$$

$\therefore$ Volume of the region $= V_1 - V_2$

$$= 12\pi - \frac{36\pi}{5}$$

$$= \frac{24\pi}{5}\ units^3$$

(d) Solve the differential equation $\frac{dy}{dx} = \frac{x+4x^2}{y^2}$, given that $y = 6$ when $x = 0$. [8]

$$\frac{dy}{dx} = \frac{x+4x^2}{y^2}$$

$$(y^2)\, dy = (x + 4x^2)\, dx$$

$$\int (y^2)\, dy = \int (x + 4x^2)\ dx$$

$$\frac{y^3}{3} = \frac{x^2}{2} + \frac{4x^3}{3} + c$$

When $x = 0, y = 6$.

So, we have,

$$\frac{(6)^3}{3} = \frac{(0)^2}{2} + \frac{4(0)^3}{3} + c$$

$$72 = 0 + 0 + c$$

$$c = 72$$

$$\therefore \frac{y^3}{3} = \frac{x^2}{2} + \frac{4x^3}{3} + 72$$

Total: 25 marks